

1. One-time pad over $Z_8 = \{0, 1, \dots, 7\}$

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HW 1

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$$C \equiv M + K \pmod{8}, \text{ key is uniform on } Z_8$$

$$\begin{aligned} \text{Message distribution: } & \Pr[M=0] = 0.5 \\ & \Pr[M=1] = 0.2 \\ & \Pr[M=2] = 0.3 \end{aligned}$$

a) Does scheme satisfy perfect security? **YES**

$$\Pr[M=m | C=c] = \Pr[M=m] \text{ for all } m, c$$

↳ i.e., seeing the cyphertext gives zero info about message

$$\Pr[M=m | C=c] = \frac{\Pr[C=c | M=m] \Pr[M=m]}{\Pr[C=c]} \text{ (Bayes thm.)}$$

$$\Pr[C=c | M=m] = \Pr[C=c] = \frac{1}{8}$$

↳ $\Pr[C=c]$ is the same for all c , thus the scheme is perfectly secured. ✓

$$\text{b) } \Pr[M=0 | C=5] = \frac{\Pr[C=5 | M=0] \Pr[M=0]}{\Pr[C=5]} \quad \text{c) } \Pr[M=1 | C=0] = \frac{\Pr[C=0 | M=1] \Pr[M=1]}{\Pr[C=0]}$$

$$\Pr[C=5 | M=0]$$

$$C \equiv 0 + K \pmod{8}$$

$$C \equiv K$$

$$= \frac{1}{8}$$

$$\Pr[C=5] = \frac{1}{8}$$

8 possibilities
1/8 satisfy $C=5$

$$\frac{(\frac{1}{8})(0.5)}{(\frac{1}{8})} = \frac{1}{2} = 0.50$$

$$\Pr[C=0 | M=1]$$

$$0 = 1 + K \pmod{8}$$

$$0 = 1 + K$$

$$K = 7 \pmod{8}$$

$$\frac{(\frac{1}{8})(\frac{1}{5})}{(\frac{1}{8})} = \frac{1}{5} = 0.20$$

$$\text{d) } \Pr[M=2 | C=7] = \frac{\Pr[C=7 | M=2] \Pr[M=2]}{\Pr[C=7]}$$

$$\Pr[C=7 | M=2] =$$

$$7 = 2 + K \pmod{8}$$

$$5 = K$$

$$\frac{1}{8} \text{ prob.}$$

$$\frac{(\frac{1}{8})(\frac{3}{10})}{(\frac{1}{8})} = \frac{3}{10} = 0.30$$

2. One-time pad over Z_{40} , $C \equiv M + K \pmod{40}$, K is uniform on Z_{40}

$$\Pr[M=10] = 0.6$$

$$\Pr[M=20] = 0.3$$

$$\Pr[M=30] = 0.1$$

a) Is scheme perfectly secure? **YES**

$$\Pr[M=m | C=c] = \Pr[M=m] \text{ for all } m, c$$

$$\begin{aligned} \Pr[C=c] &= \sum \frac{1}{40} (\Pr[M=m]) \\ &= \frac{1}{40} (.6 + .3 + .1) = \frac{1}{40} \end{aligned}$$

$$\Pr[M=m | C=c] = \frac{\frac{1}{40} (\Pr[M=m])}{\frac{1}{40}} = \Pr[M=m]$$

The scheme is perfectly secure ✓
because $\Pr[M=m | C=c] = \Pr[M=m]$

$$\text{b) } \Pr[M=10 | C=15] = \frac{\Pr[C=15 | M=10] \Pr[M=10]}{\Pr[C=15]}$$

$$\Pr[C=15 | M=10]$$

$$\Pr[M=10] = 0.6$$

$$\hookrightarrow 15 = 10 + K \pmod{40} \quad \Pr[C=15] = \frac{1}{40}$$

$$5 = K \rightarrow \frac{1}{40}$$

$$\frac{(\frac{1}{40})(\frac{3}{5})}{(\frac{1}{40})} = \frac{3}{5} = 0.60$$

$$c) \Pr[M=20|C=25] = \frac{\Pr[C=25|M=20] \Pr[M=20]}{\Pr[C=25]}$$

$$\Pr[C=25|M=20]$$

$$25 = 20 + K \pmod{40}$$

$$5 = K$$

$$\hookrightarrow 1/40 \text{ prob.}$$

$$\frac{(1/40)(3/10)}{(1/40)} = \frac{3}{10} = 0.30$$

$$M=20 \rightarrow 3/10$$

$$\Pr[C=25] = 1/40$$

$$d) \Pr[M=30|C=5] = \frac{\Pr[C=5|M=30] \Pr[M=30]}{\Pr[C=5]}$$

$$\Pr[C=5|M=30]$$

$$5 = 30 + K \pmod{40}$$

$$15 = K \rightarrow 1/40$$

$$\Pr[M=30] = 0.1 + 1/10$$

$$\frac{(1/40)(1/10)}{(1/40)} = \frac{1}{10} = 0.10$$

3. consider $\mathbb{Z}_6$: $C \equiv M + 2K \pmod{6}$, K is uniform
 $2K \pmod{6} = \{0, 2, 4\}$ uniformly $\rightarrow 1/3$ prob. of each

$$\Pr[M=0] = 0.2$$

$$\Pr[M=1] = 0.1$$

$$\Pr[M=2] = 0.4$$

$$\Pr[M=4] = 0.3$$

a) Does the scheme satisfy perfect security? **NO**

$$C \equiv M + \{0, 2, 4\} \pmod{6}$$

IF $M=0$:

$$C = \{0, 2, 4\}$$

IF $M=1$:

$$C = 1 + \{0, 2, 4\} \pmod{6}$$

$$C = \{1, 3, 5\}$$

... same for

$$M=2 \text{ and } M=4$$

$$\hookrightarrow C = \{0, 2, 4\}$$

$\hookrightarrow$ IF the cypher is even, the message is even
 IF odd, the message is odd.

we can deduce info about the message from the cypher, $\therefore$ the cypher is **not perfectly secure**.

$$b) \Pr[M=0|C=4] = \frac{\Pr[C=4|M=0] \Pr[M=0]}{\Pr[C=4]} \rightarrow \frac{(1/3)(1/5)}{3/10} = \left(\frac{1}{15}\right) \frac{10}{3} = \frac{10}{45} = 0.22$$

$$\Pr[C=4|M=0]$$

$$4 = M + \{0, 2, 4\} \pmod{6}$$

$$4 = \{0, 2, 4\} \pmod{6}$$

$$\hookrightarrow 1/3$$

$$\Pr[C=4] = \frac{1}{3}(0.2 + .4 + .3) = \frac{1}{3}(0.9) = \frac{3}{10}$$

(can only be produced
 by $m = 0, 2, 4$)

$$c) \Pr[M=1 | C=3] = \frac{\Pr[C=3 | M=1] \Pr[M=1]}{\Pr[C=3]} = \frac{(1/3)(1/10)}{(1/30)} = \left(\frac{1}{30}\right) \frac{30}{1} = 1$$

$$\Pr[C=3 | M=1]$$

$$\Pr[M=1] = 0.1$$

$$3 = 1 + \{0, 2, 4\} \% 6$$

$$2 = \{0, 2, 4\} \% 6$$

$$= 1/3$$

$\Pr[C=3] \rightarrow$ can only be

generated by $M=1$ ($\frac{1}{3}$)

$$\Pr[C=3] = \left(\frac{1}{3}\right)\left(\frac{1}{10}\right) = \frac{1}{30}$$

$$d) \Pr[M=2 | C=0] = \frac{\Pr[C=0 | M=2] \Pr[M=2]}{\Pr[C=0]}$$

$$\Pr[C=0 | M=2]$$

$$\hookrightarrow 0 \equiv 2 + \{0, 2, 4\} \% 6$$

$$\hookrightarrow 1/3$$

$$= \frac{\left(\frac{1}{3}\right)\left(\frac{2}{5}\right)}{(3/10)} = \frac{2}{15} \left(\frac{10}{3}\right) = \frac{20}{45} = 0.44$$

$\Pr[C=0] \rightarrow$ can be generated

by even m values

$$(M=0, 2, 4)$$

$$\frac{1}{3}(0.2 + 0.4 + 0.3) = \frac{.2}{3} + \frac{.4}{3} + \frac{.3}{3} = \frac{.9}{3} = .3$$

$$\Pr[M=2] = 0.4$$

4. Scheme over $\mathbb{Z}_6$ using $C \equiv M + K \pmod{6}$,

key distribution is biased: , Message distribution:

$$\Pr[K=0] = 0.1$$

$$\Pr[M=0] = 0.2$$

$$\Pr[K=1] = 0.3$$

$$\Pr[M=1] = 0.3$$

$$\Pr[K=2] = 0.6$$

$$\Pr[M=2] = 0.5$$

$\hookrightarrow 0$ for other K

a. Does this scheme satisfy perfect security? **NO**

$$\Pr[M=m | C=c] = \Pr[M=m] \text{ for all } m, c$$

$$M \in \{0, 1, 2\}, K \in \{0, 1, 2\}$$

$\hookrightarrow$ pick any cypher text and check above equation:

$$C \in \{0, 1, 2, 3, 4\}$$

$$\text{let } C=2, C \equiv M + K \pmod{6}$$

$$2 = M + K$$

$$M = 2 - K$$

$\downarrow$ possible keys

$$K=0, M=2$$

$$K=1, M=1$$

$$K=2, M=0$$

$$\Pr[C=2] = \sum \Pr[M=2-K] \Pr[K=K]$$

$$\Pr[C=2] = (0.1)(0.5) + (0.3)(0.3) + (0.6)(0.2)$$

$$\Pr[C=2] = 0.05 + 0.09 + 0.12 = 0.26$$

$$\Pr[M=0 | C=2] = \frac{\Pr[M=0 | C=2]}{\Pr[C=2]} = \frac{(0.2)(0.6)}{0.26} = 0.462$$

$\downarrow$
 $\Pr[M=0] = 0.2 \neq 0.462 \therefore$ NOT Perfectly secure

$$b) \Pr[M=0 | C=2] = \frac{\Pr(M=0) \cap \Pr(C=2)}{\Pr[C=2]} = \frac{0.12}{0.26} = 0.46$$

$$\Pr[C=2 | M=0] =$$

$$C \equiv M + K \pmod{6}$$

$$2 \equiv 0 + K \pmod{6}$$

$$\downarrow$$

$$2 \equiv K$$

cypher text is 2 iff

$K=2$ and $M=0$

$$= \Pr(K=2) \cdot \Pr(M=0)$$

$$= (0.6)(0.2) = 0.12$$

$$\Pr[C=2]$$

$$2 = M + K$$

$\hookrightarrow (M, K)$ pairs that

work =

$$(0, 2), (2, 0), (1, 1)$$

$$\Pr(C=2) = (0.2)(0.6) + (0.5)(0.1) + (0.3)(0.3)$$

$$= 0.12 + 0.05 + 0.09 = 0.26$$

$$\Pr(M=0) = 0.2 \text{ (given)}$$

$$c) \Pr[M=1 | C=3] = \frac{\Pr(M=1) \cdot \Pr(C=3)}{\Pr(C=3)} = \frac{0.18}{0.33} = 0.545 = 0.55$$

$$\Pr[M=1] \cdot \Pr[C=3]$$

$\downarrow$ Find possible pairs (M, K)

$$C \equiv M + K \pmod{6}$$

$$3 \equiv 1 + K$$

$$2 \equiv K$$

$\downarrow K=2$ and $M=1$:

$$= (0.6)(0.3) = 0.18$$

$$\Pr(C=3)$$

$$3 = M + K$$

(M, K) pairs

$$\hookrightarrow (1, 2), (2, 1)$$

$$\Pr(C=3) = (0.3)(0.6) + (0.5)(0.3)$$

$$= 0.18 + 0.15 = 0.33$$

$$d) \Pr[M=0 | C=1] = \frac{\Pr[C=1] \cap \Pr[M=0]}{\Pr[C=1]} = \frac{0.06}{0.09} = 0.67$$

$$\Pr[C=1] \cap \Pr[M=0]$$

$$C \equiv M + K \pmod{6}$$

$$1 \equiv K \pmod{6}$$

$\hookrightarrow$ pair $(0, 1)$ are valid for C

$$= \Pr(K=1) \cdot \Pr(M=0)$$

$$= (0.3)(0.2) = 0.06$$

$$\Pr[C=1]$$

$$1 = M + K$$

(M, K) pairs:

$$(0, 1), (1, 0)$$

$$\Pr(C=1) = (0.1)(0.3) + (0.3)(0.2)$$

$$= 0.03 + 0.06$$

$$= 0.09$$

5) Scheme over $\mathbb{Z}_6$ using $C \equiv M + K \pmod{6}$, key is chosen uniformly

$$\Pr[K=0, 1, 2] = 1/3$$

$$K \neq 3, 4, 5$$

$$\Pr[M=1] = 0.2$$

$$\Pr[M=2] = 0.3$$

$$\Pr[M=3] = 0.5$$

$$C \in \{1, 2, 3, 4, 5\}$$

(M, K)	C	$\Pr(M), \Pr(K)$
(1, 0)	1	(0.2, 1/3)
(1, 1)	2	(0.2, 1/3)
(1, 2)	3	(0.2, 1/3)
(2, 0)	2	(0.3, 1/3)
(2, 1)	3	(0.3, 1/3)
(2, 2)	4	(0.3, 1/3)
(3, 0)	3	(0.5, 1/3)
(3, 1)	4	(0.5, 1/3)
(3, 2)	5	(0.5, 1/3)

a. is the encryption scheme perfectly secure?

$$\Pr[M=m | C=c] = \Pr[M=m] \quad \forall M, C ? \quad \text{NO}$$

$$\Pr[M=m | C=c] = \frac{\Pr[C=c | M=m] \cdot \Pr[M=m]}{\Pr[C=c]} = \frac{\frac{2/30}{1/3} = \frac{6}{30} = \frac{1}{5} = 0.2}{\downarrow}$$

test pairs where $C=3 : (1,2), (2,1), (3,0)$

$$\Pr[C=3] = (0.3)(\frac{1}{3}) + (0.3)(\frac{1}{3}) + (0.5)(\frac{1}{3})$$

$$\Pr[C=3] = 1/3$$

$$\Pr[M=1] = 0.2$$

this pair is
secure, need to
check more

$$\Pr[M=1, C=3] = \Pr[M=1, K=2] = \Pr[M=1] \Pr[K=2] = (0.2)(\frac{1}{3}) = \frac{2}{30}$$

let $M=3$,

$$M \quad K \quad = \quad C$$

$$3 + 0 = 3$$

$$3 + 1 = 4$$

$$3 + 2 = 5$$

$$\hookrightarrow C \text{ can only } = 3-5 \quad (\Pr[M=3 | C=2] = 0)$$

$$\Pr[M=3] = 0.5 \uparrow$$

$$b) \Pr[M=1 | C=3] = \frac{0.2(\frac{1}{3})}{1/3} = 0.2$$

the probability of $M=3$
changes depending on the
cypher text

$\therefore$ NOT SECURE

(M,K) pairs that $C=3$

$$\hookrightarrow (1,2), (2,1), (3,0)$$

$$\text{prob each} : (0.2)(\frac{1}{3}), (0.3)(\frac{1}{3}), (0.5)(\frac{1}{3}) = 1/3$$

$$c) \Pr[M=2 | C=4] \rightarrow \frac{.10}{.266} = .37$$

(M,K) pairs that $C=4$

$$\hookrightarrow (2,2), (3,1)$$

$$\text{prob ea} : \underbrace{(0.3)(\frac{1}{3})}_{.10} + \underbrace{(0.5)(\frac{1}{3})}_{.16} = 0.266$$

$$d) \Pr[M=1 | C=2] \rightarrow \frac{0.2}{3} = \frac{0.2}{\frac{0.2}{3} + \frac{0.3}{3}} = \frac{0.2}{0.5} = .2 = 0.39$$

(M,K) pairs that $C=2 :$

$$(1,1) (2,0)$$

$$\text{prob ea} : (0.2)(\frac{1}{3}) (0.3)(\frac{1}{3})$$

$$\Pr[C=2] = (0.2)(\frac{1}{3}) + (0.3)(\frac{1}{3}) = 0.66 + 0.1 = .76$$